Arunabha Majumder



Mechanical Design Engineer

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PROFILE

Mechanical design engineer with a fine arts background. I turn concepts and sketches into parametric CAD parts and assemblies (SolidWorks), 3D print them, assemble them and fit test them on real hardware. Most of my work is wearable robotics: I designed a shoulder exoskeleton that sits on the human body, with a motor, encoder and sensors built into it.

EXPERIENCE

PhD Researcher, Robotics (project VIEXO)
Aalborg University, Denmark | *2023 - Present*

- Designed a hybrid shoulder exoskeleton from concept to prototype: a variable stiffness mechanism with a worm gear for pretension, working in parallel with a 6 Nm motor. It uses more than 25% less motor energy (Best Paper Award, IFToMM ISRM 2026).
- Built the full parametric CAD assembly and technical drawings in SolidWorks: parts, mounting points, fasteners and clearances around the moving joints; optimised the mechanism with a genetic algorithm.
- 3D printed parts for fit and function testing, refined them over several iterations, then had the final parts manufactured.
- Integrated the motor, encoder and torque/force sensors into the structure; measured the stiffness on a physical test setup.
- Design for manufacture and assembly: replaced a clutch part that kept breaking and was hard to make with a simple bolt-locking clutch (8 locking angles) that was easy to make, fit and replace.
- Designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM); estimated its stiffness with less than 10% error and quantified human arm impedance.
- Taught university courses and supervised Bachelor's students (2023 - 2025).

MTech Researcher, Soft Exosuit (Pneumatic Muscles)
CSIR-CMERI, India | *2020 - 2022*

- Fabricated and assembled pneumatic artificial muscles for a soft exosuit; designed the outer PID loop of a neural network-based controller in MATLAB/Simulink and implemented it in LabVIEW (IEEE CONECCT 2022).

Bachelor's Capstone and Student Teams
Siddaganga Institute of Technology, India | *2017 - 2019*

- Designed and built an arm exoskeleton and exo glove: CAD, machining, assembly and testing (TATA Technologies Prayatna Innovation Fund grant; Best Major Project Award).
- Built a Go-Kart (2017) and an Efficycle hybrid vehicle (2018) with student teams.

SKILLS

CAD: SolidWorks (CSWA): parametric parts and assemblies, technical drawings, tolerances, fasteners, clearances, organised assembly files

Prototyping: 3D printing (Bambu Lab, Ultimaker Cura), design for 3D printing, fit testing, machining and hands-on assembly, DFM and DFA

Integration: motors, encoders, torque/force sensors, Arduino, MATLAB/Simulink

Art and graphic design: hand sketching of concepts, painting, technical illustration and visuals for papers, posters and demos; Adobe Illustrator, Photoshop, Procreate

EDUCATION

PhD, Mechatronics and Robotics, Aalborg University, Denmark | 2023 - present (research complete)

MTech, Mechatronics, AcSIR - CSIR-CMERI, India | 2020 - 2022

BTech, Mechanical Engineering, Siddaganga Institute of Technology, India | 2015 - 2019

Senior Diploma in Fine Arts (Painting), Bangiya Sangeet Parishad | 2008, First Division with Distinction

LANGUAGES AND REFERENCES

English (C1), Bengali (native), Hindi (spoken)

References: Prof. Shaoping Bai (sbh@mp.aau.dk); Assoc. Prof. Lars Rosgaard Jensen; Assoc. Prof. Anderson de Souza Castelo Oliveira; all Department of Materials and Production, Aalborg University